

Hydra Settlement: A Synchronous Layer-2 for eUTXO Chains

Working Paper, Test Fixture Edition

1. Introduction

Hydra is a family of Layer-2 protocols designed to scale Cardano-style eUTXO ledgers by allowing a small set of participants to run an isolated, high-throughput state channel called a Hydra Head. Transactions inside a Head settle instantly among participants and are only reconciled with the main chain when the Head is closed.

2. Architecture

2.1 Head Lifecycle

A Hydra Head moves through four phases: Init, Open, Close, and Fanout. During Init, participants commit UTXOs from the main chain. During Open, participants exchange signed snapshots off-chain. Close posts the latest snapshot on-chain, and Fanout distributes the final UTXO set back to the main ledger.

2.2 Settlement Guarantees

Because every state transition inside the Head requires unanimous signatures from all participants, Hydra can be used as a synchronous settlement layer over Cardano-style eUTXO state: once all parties sign a snapshot, that state is final among them even before it touches the main chain.

3. Throughput Comparison

The table below compares theoretical transaction throughput across configurations tested in the reference implementation.

Configuration	Participants	TPS	Finality
Baseline L1	-	250	~20s
Hydra Head (small)	3	1,000	<1s
Hydra Head (large)	10	800	<1s

4. Threat Model

Hydra Heads assume an honest majority is not required; instead, safety relies on unanimity and a contestation period during Close. If a participant posts a stale snapshot, any other participant can contest it on-chain within the contestation window by presenting a newer, validly signed snapshot.

5. Open Questions

It remains an open question how Hydra Heads compose with external delegated-signing standards such as the Open Wallet Standard (OWS), particularly when an autonomous agent — rather than a human participant — holds one of the Head's signing keys.

Figure 1: Head Lifecycle (placeholder)